# Markets

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# Chapter 5: Price Controls and Quotas

# Market Interventions

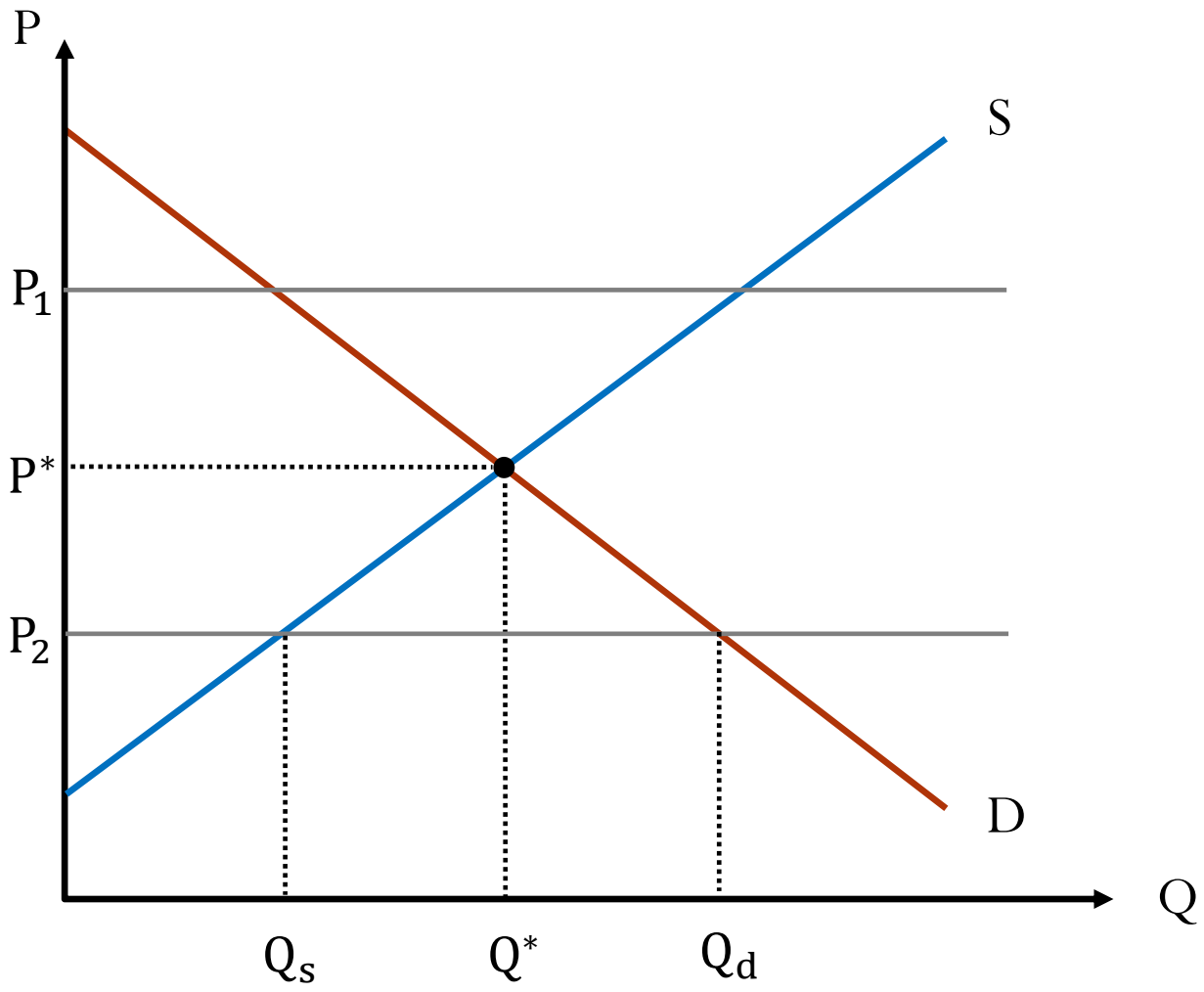
- Market interventions on either prices or quantities distorts the economic signals that guide consumers and producers.
- These distorted prices often provide poor information, leading to inefficient decisions and outcomes.
- Price controls: legal restrictions on market price.
  - Price ceiling (e.g., rent control) vs. Price floor (e.g., minimum wage)
- Quantity control: legal restrictions on total transactions (e.g., license quota)

# Price Ceilings

# Price Ceilings

- Non-binding price ceiling: $P_1 > P^*$
- Binding price ceiling: $P_2 < P^*$
- Shortage caused by price ceiling:

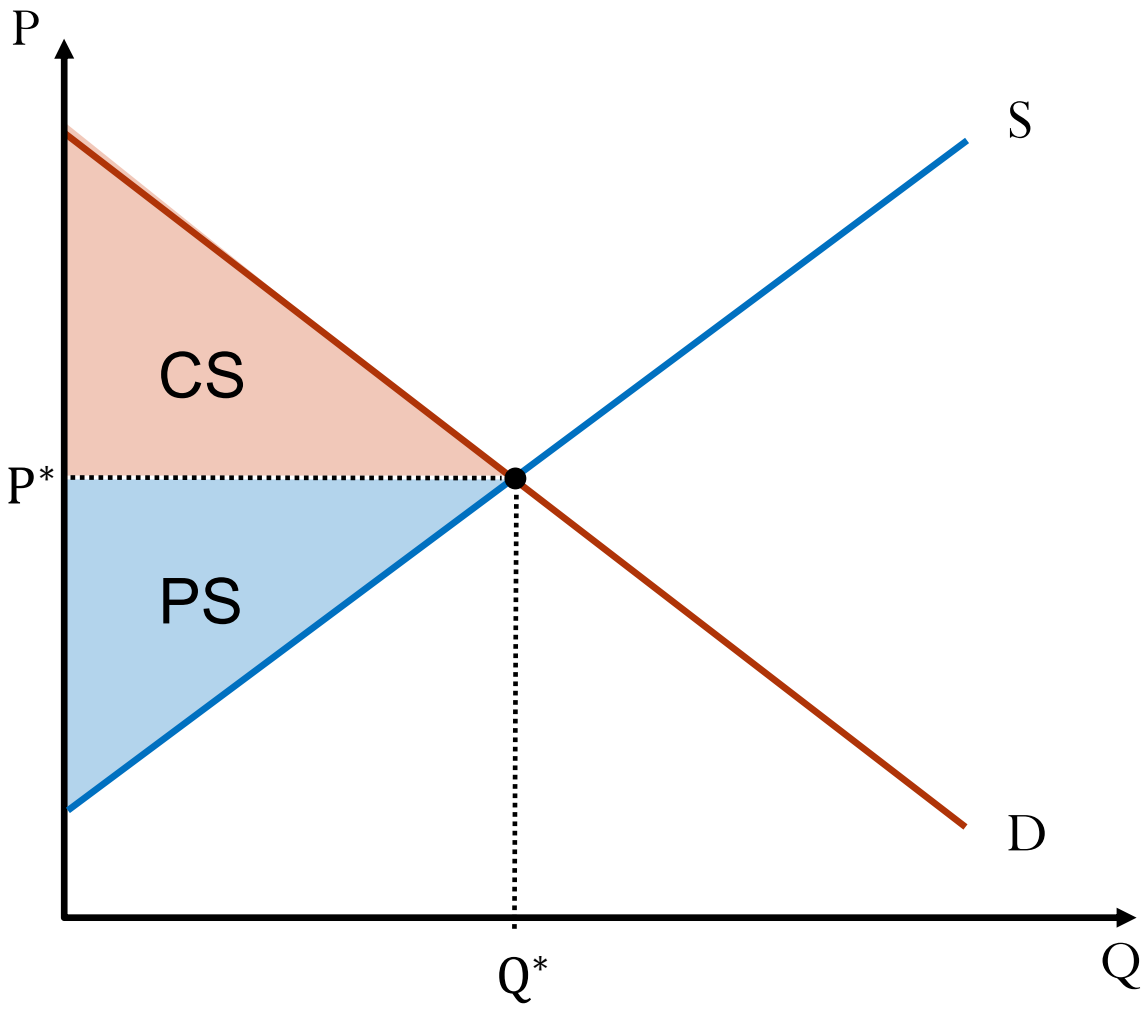
$$Q_d - Q_s$$

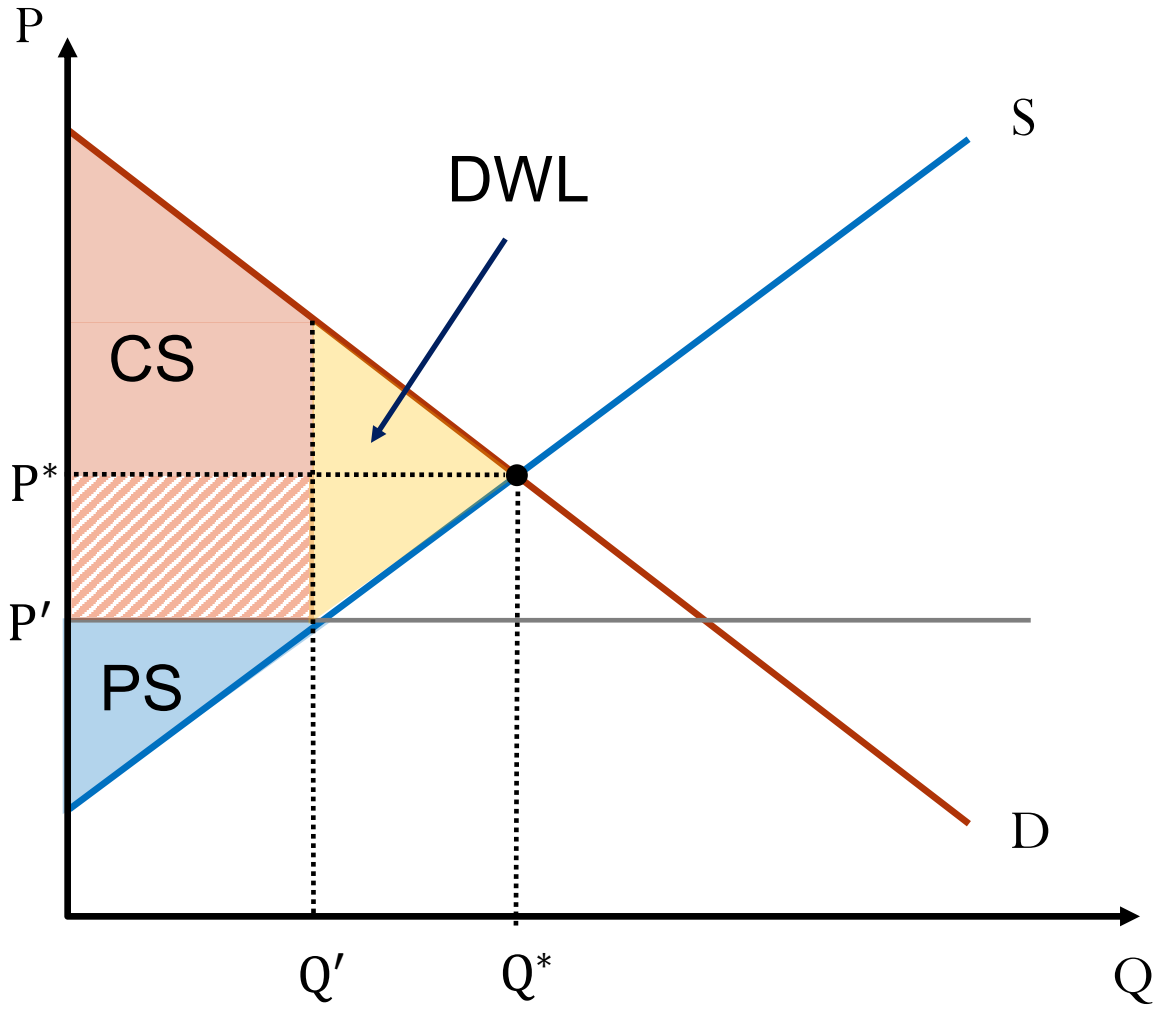


# How Price Ceilings Cause Inefficiency?

- Inefficiently low quantity → **deadweight loss** (loss of total surplus)
- Inefficient allocation to consumers
- Wasted resources
- Inefficiently low quality
- Black markets

# Deadweight Loss

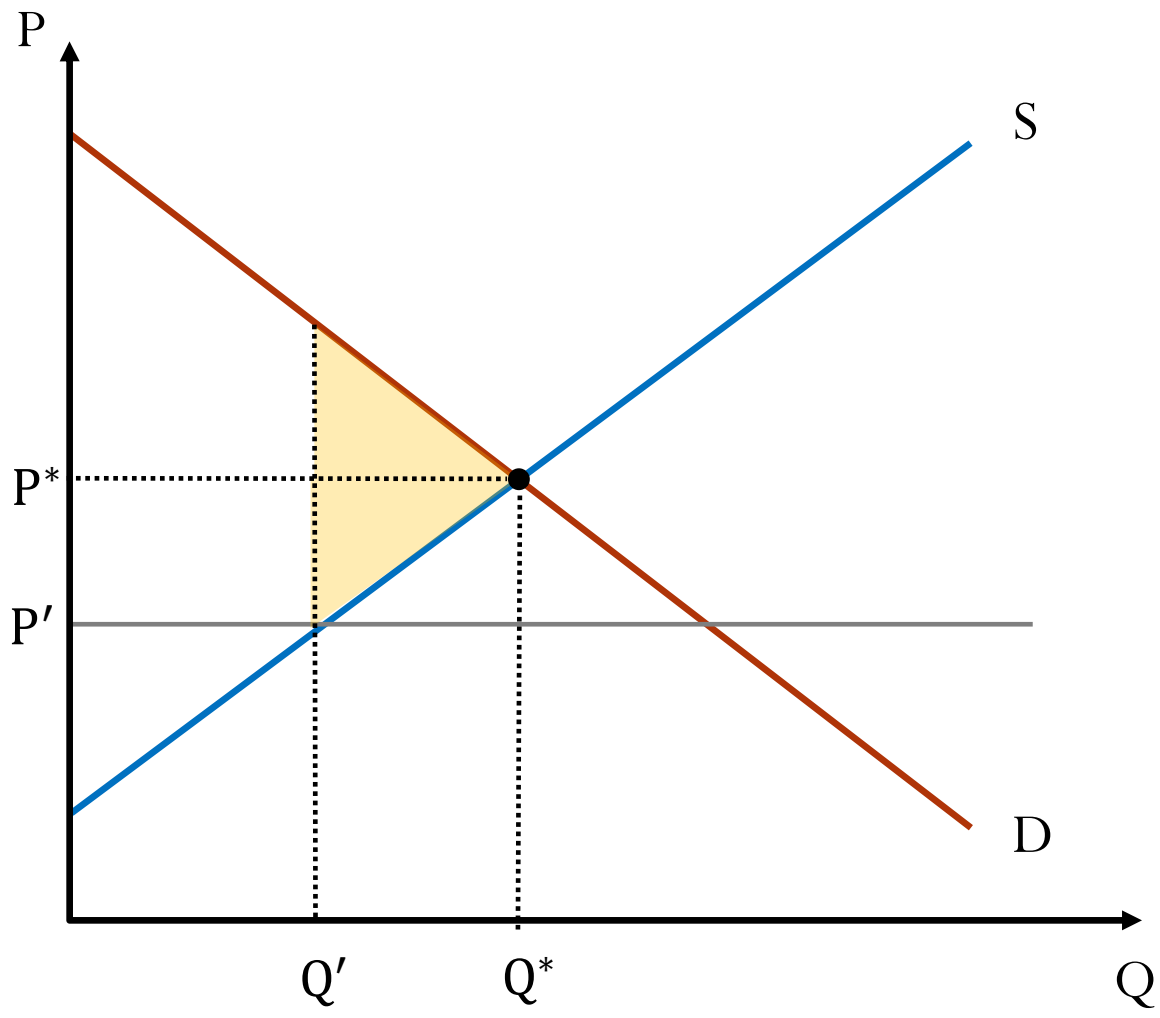


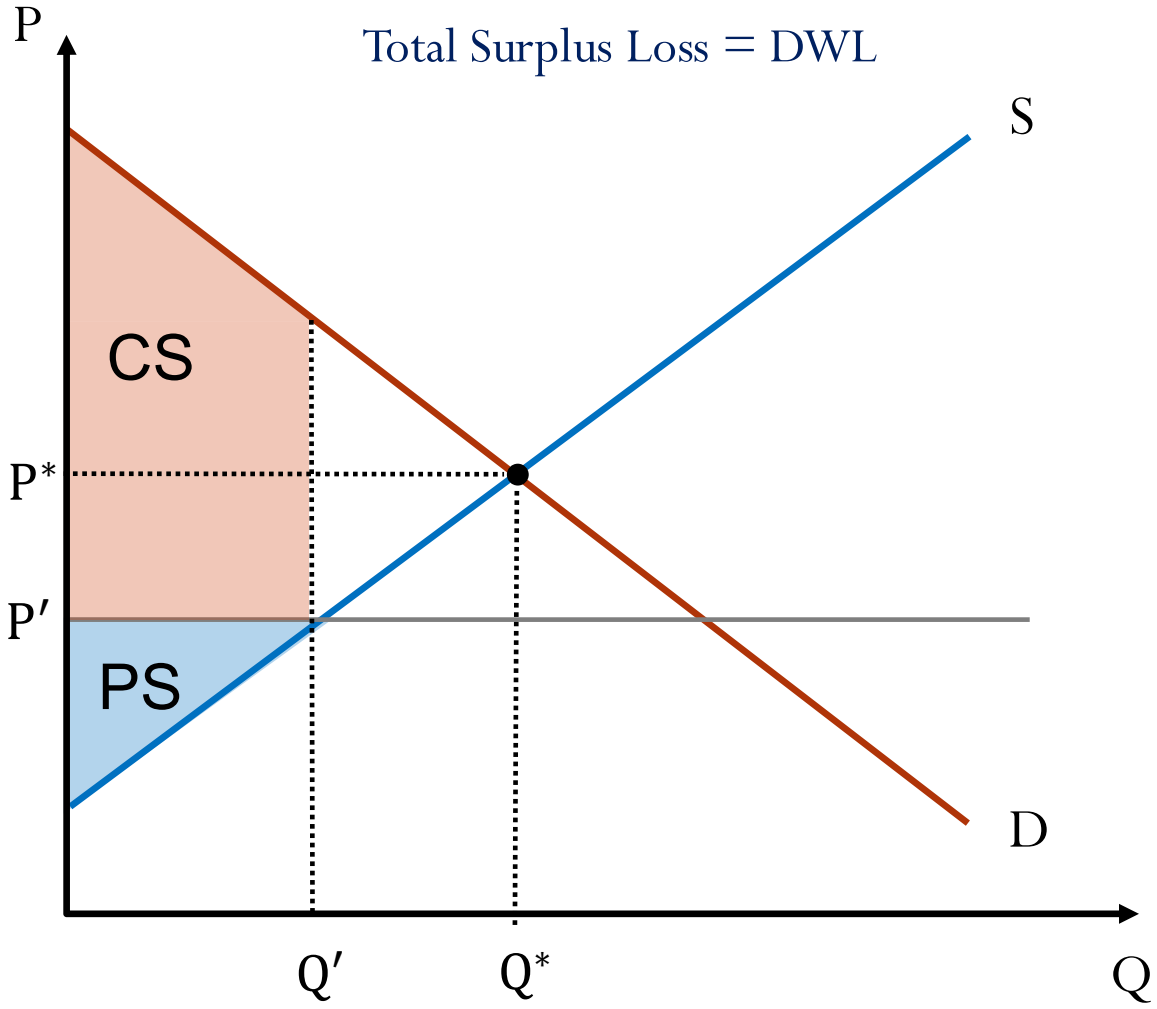
# Deadweight Loss

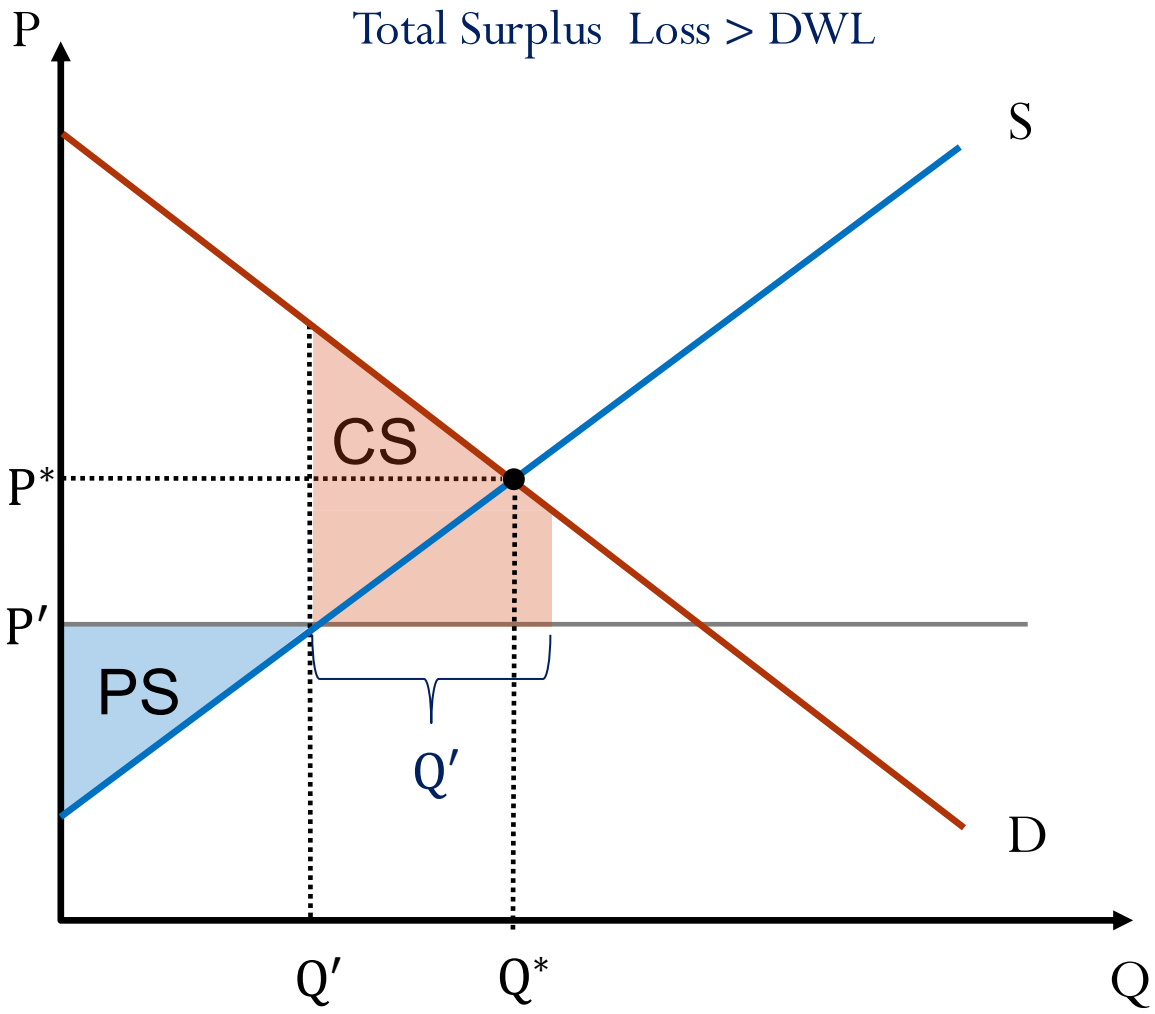
## Deadweight Loss

The reduction in a society's total surplus that results from a market distortion.



# Inefficient Allocation



# Other Predictable Inefficiency

- How would people compete if they cannot offer a higher price?
- How would producer react if they cannot raise price?
- Effects of black markets?
- Why do price ceilings exist and may persist?

# The Effects of Rent Control Expansion on Tenants, Landlords, and Inequality: Evidence from San Francisco†

*By* Rebecca Diamond, Tim McQuade, and Franklin Qian*

*Using a 1994 law change, we exploit quasi-experimental variation in the assignment of rent control in San Francisco to study its impacts on tenants and landlords. Leveraging new data tracking individuals' migration, we find rent control limits renters' mobility by 20 percent and lowers displacement from San Francisco. Landlords treated by rent control reduce rental housing supplies by 15 percent by selling to owner-occupants and redeveloping buildings. Thus, while rent control prevents displacement of incumbent renters in the short run, the lost rental housing supply likely drove up market rents in the long run, ultimately undermining the goals of the law.* (*JEL* R23, R31, R38)

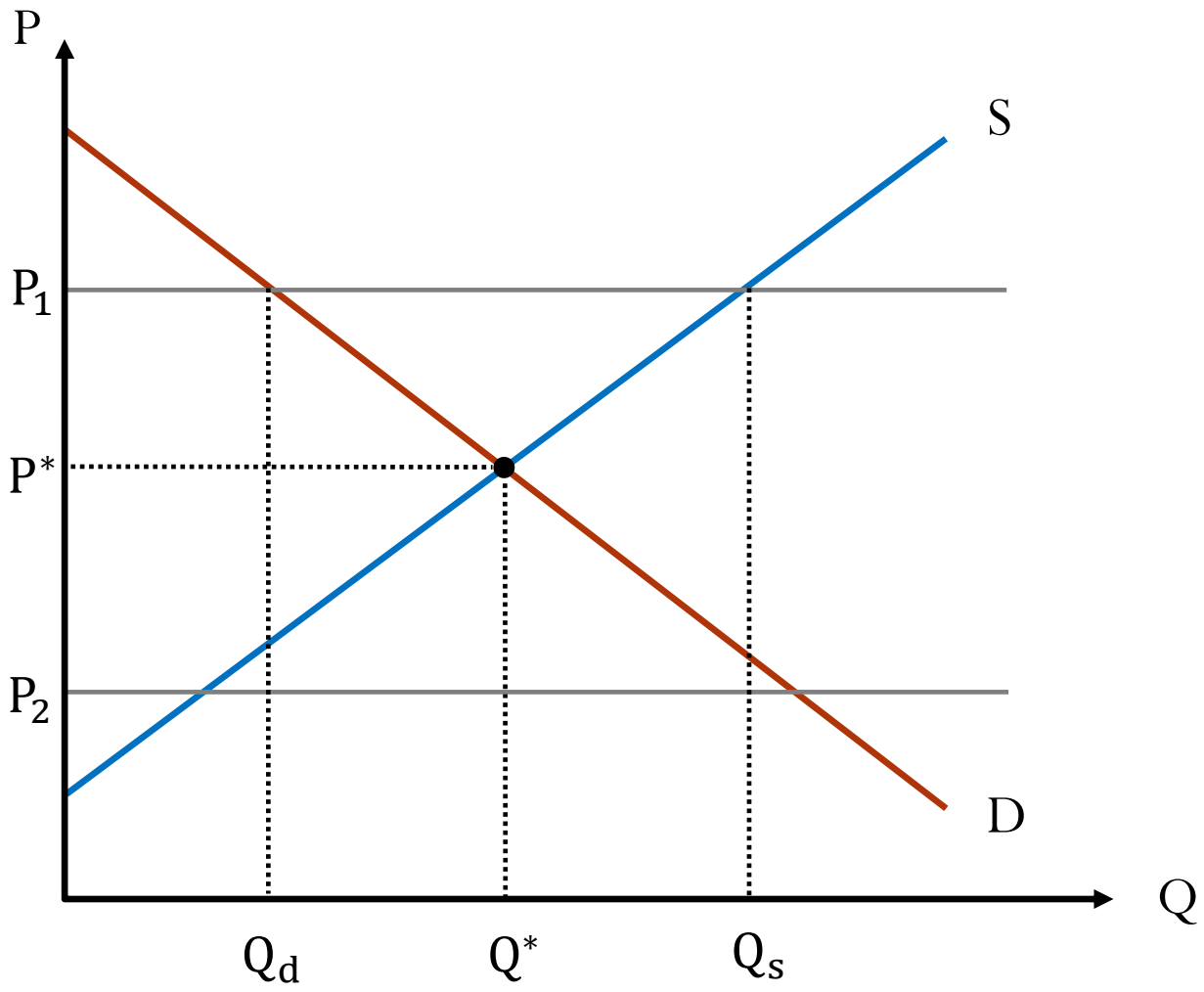
Figure 3. Geographic Distribution of Treated and Control Buildings in San Francisco

# Price Floors

# Price Floors

- Non-binding price floor: $P_2 < P^*$
- Binding price floor: $P_1 > P^*$
- Excess supply caused by price floor:

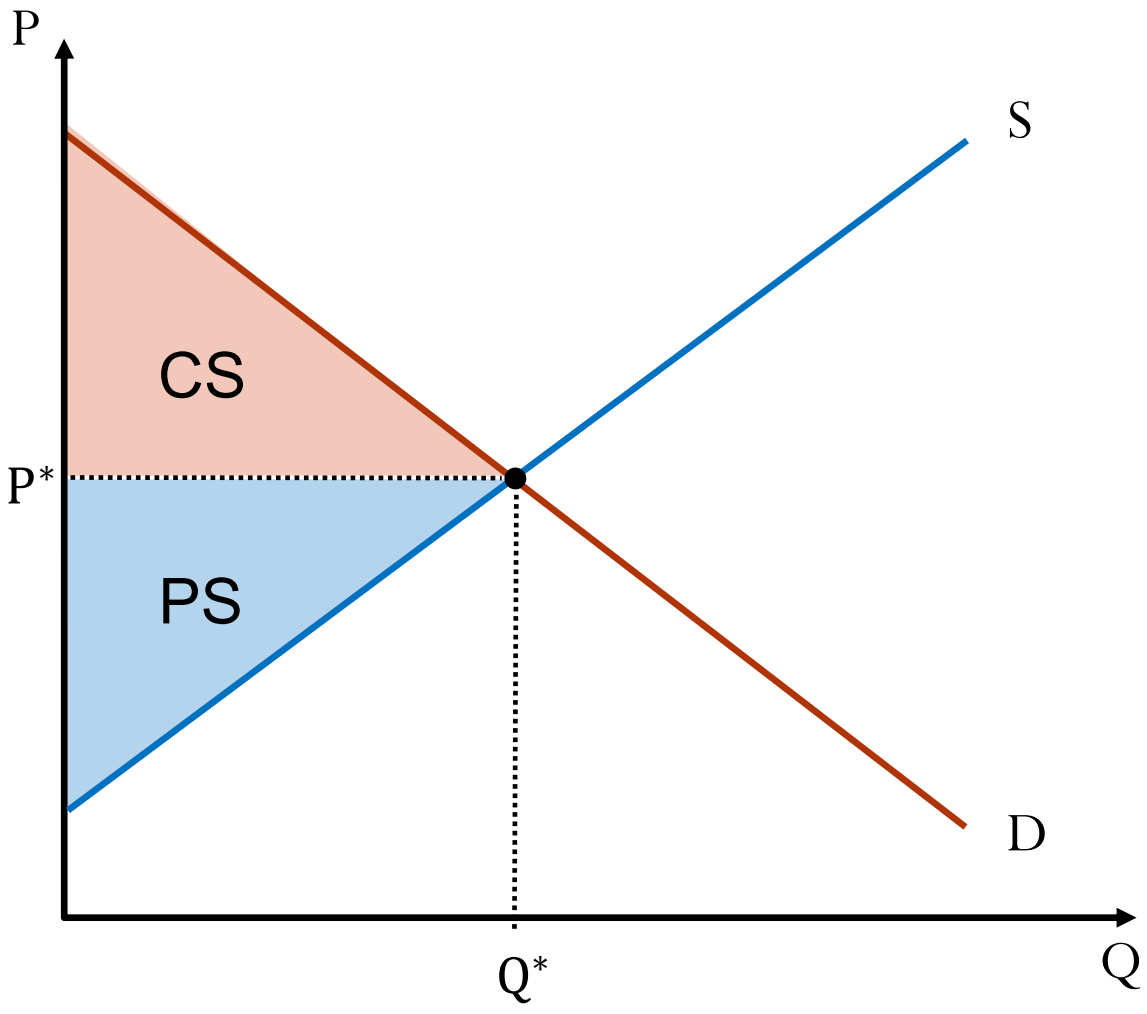
$$Q_s - Q_d$$

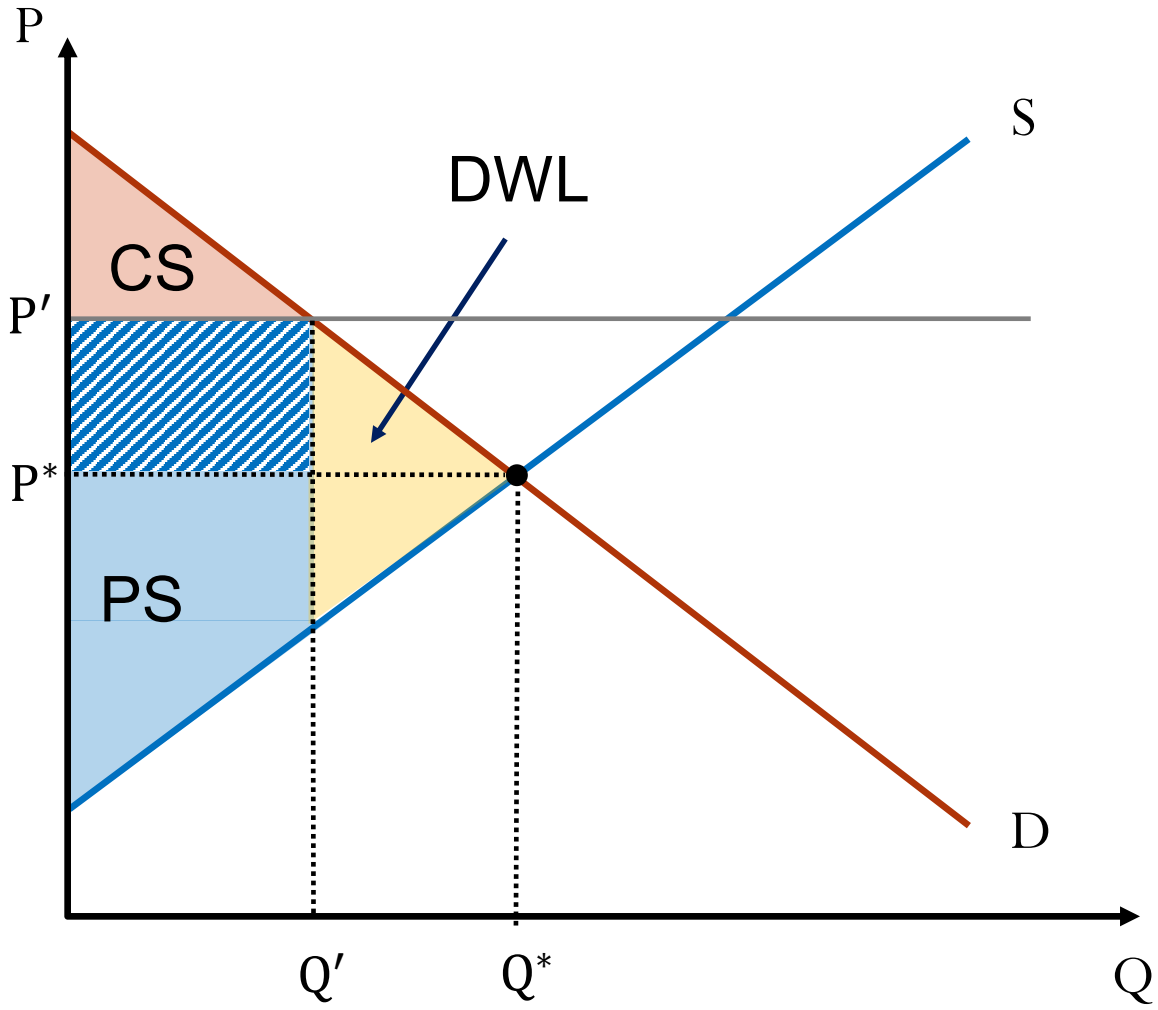


# How Price Floors Cause Inefficiency?

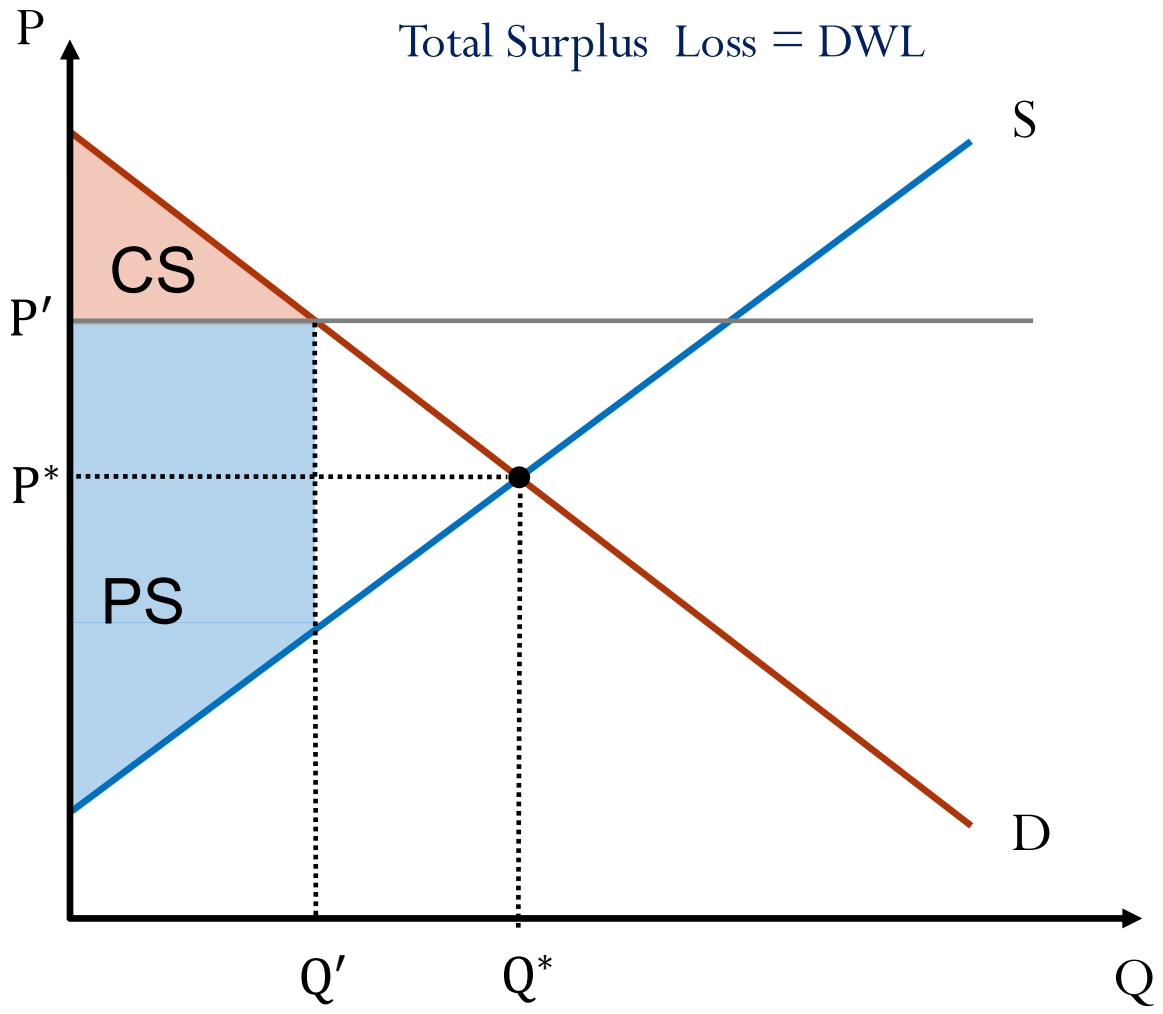
- Inefficiently low quantity → deadweight loss (loss of total surplus)
- Inefficient allocation to producers
- Wasted resources
- Inefficiently high quality
- Black markets

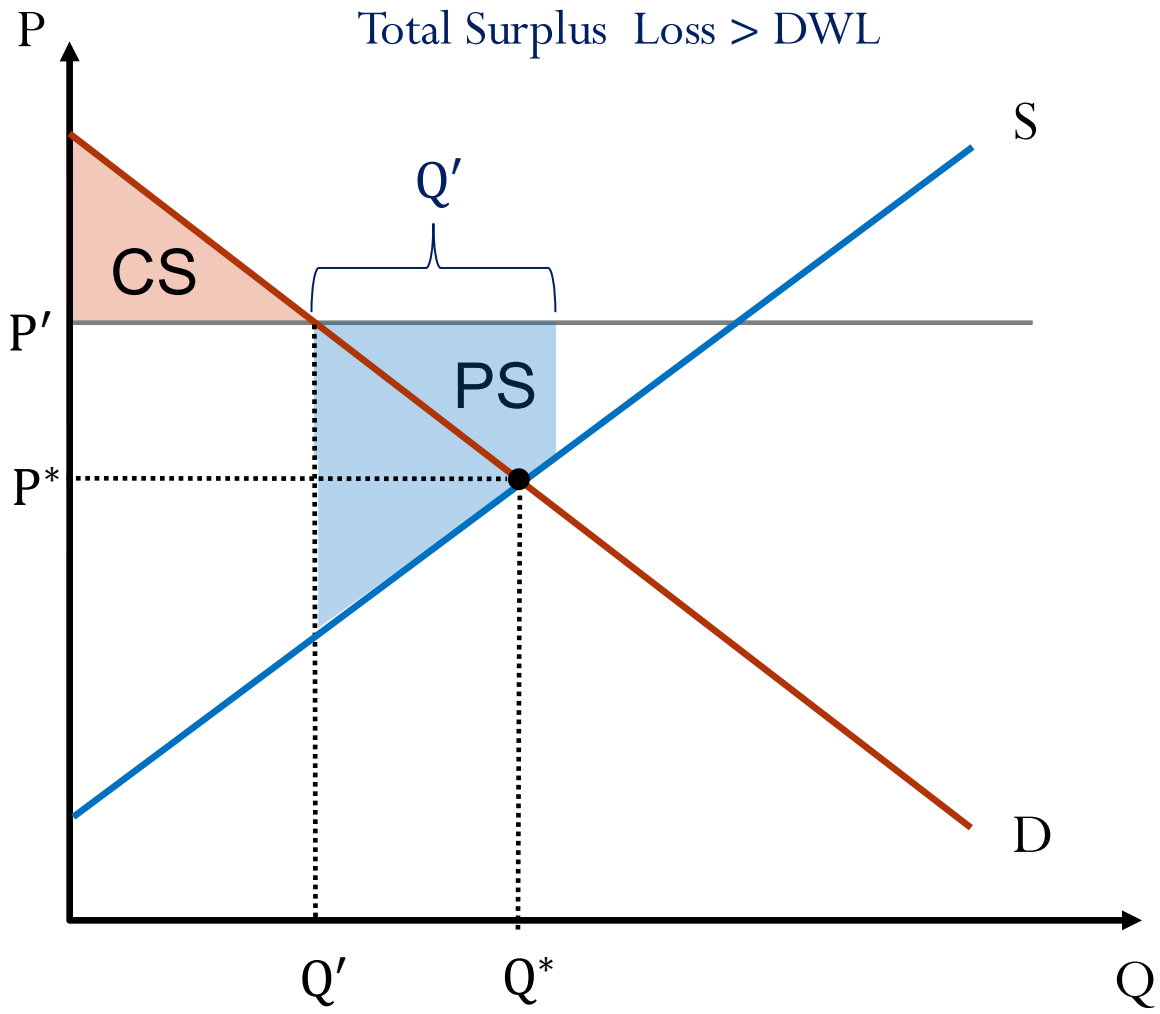
# Deadweight Loss



# Inefficient Allocation



# Other Predictable Inefficiency

- Direct consequence of over-supply?
- How would producer compete if they cannot lower price?
- Effect of black markets?
- Why do price floors exist and may persist?

# Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania

*By* David Card and Alan B. Krueger*

*On April 1, 1992, New Jersey's minimum wage rose from $4.25 to $5.05 per hour. To evaluate the impact of the law we surveyed 410 fast-food restaurants in New Jersey and eastern Pennsylvania before and after the rise. Comparisons of employment growth at stores in New Jersey and Pennsylvania (where the minimum wage was constant) provide simple estimates of the effect of the higher minimum wage. We also compare employment changes at stores in New Jersey that were initially paying high wages (above $5) to the changes at lower-wage stores. We find no indication that the rise in the minimum wage reduced employment.* (*JEL* J30, J23)

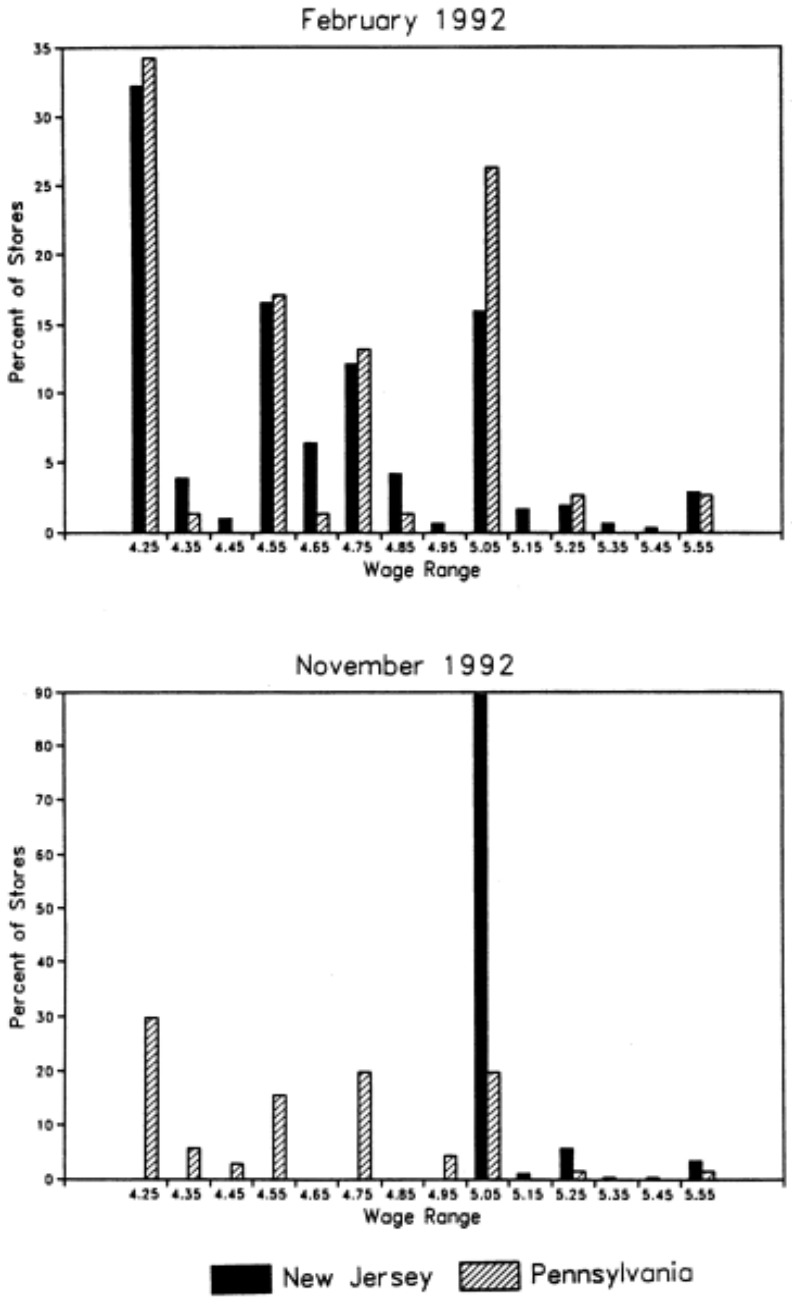



Figure 1. Distribution of Starting Wage Rates

# Quantity Control

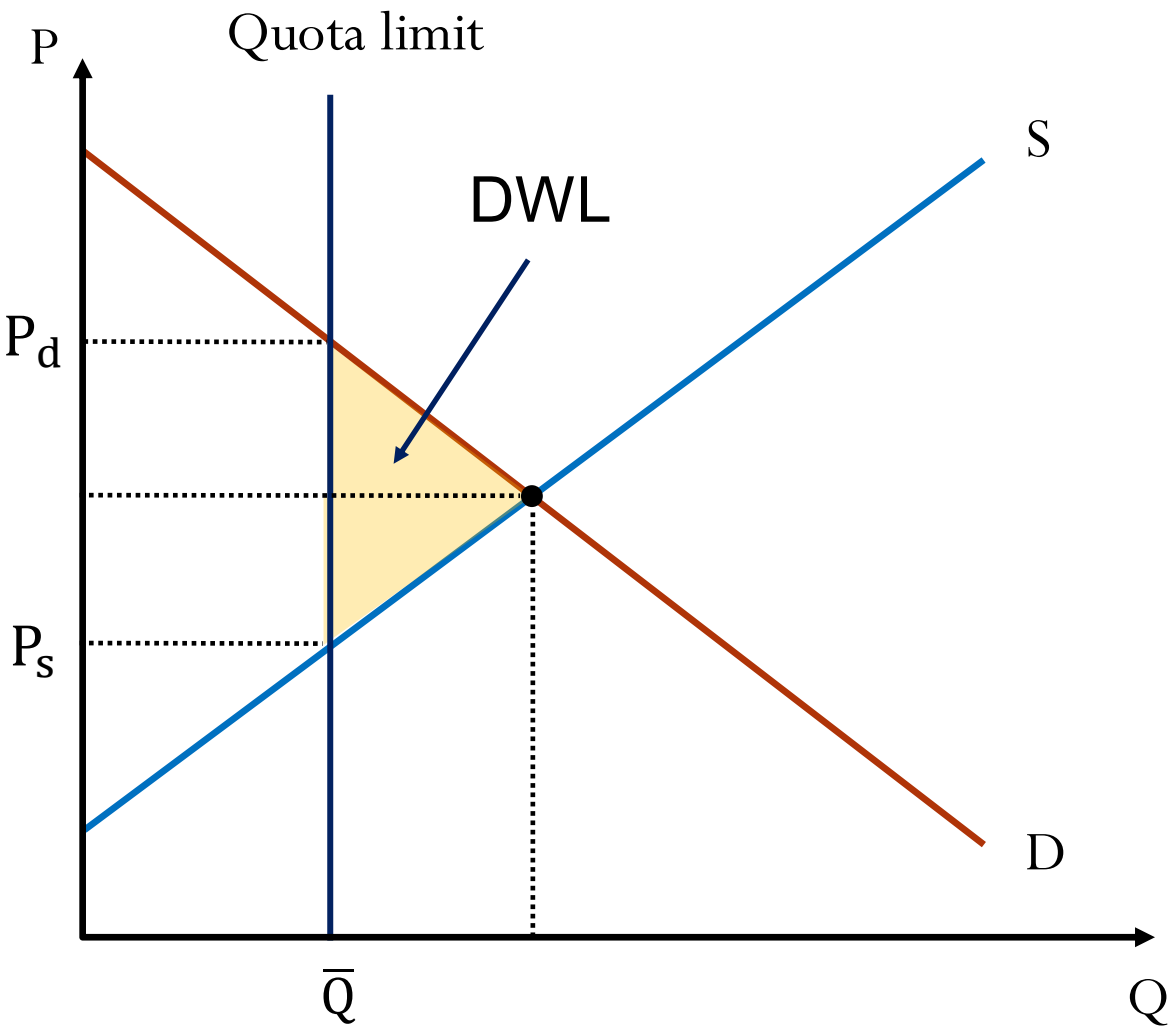
# Quantity Control

## Demand Price

The price at which a specific quantity is demanded by consumers.

## Supply Price

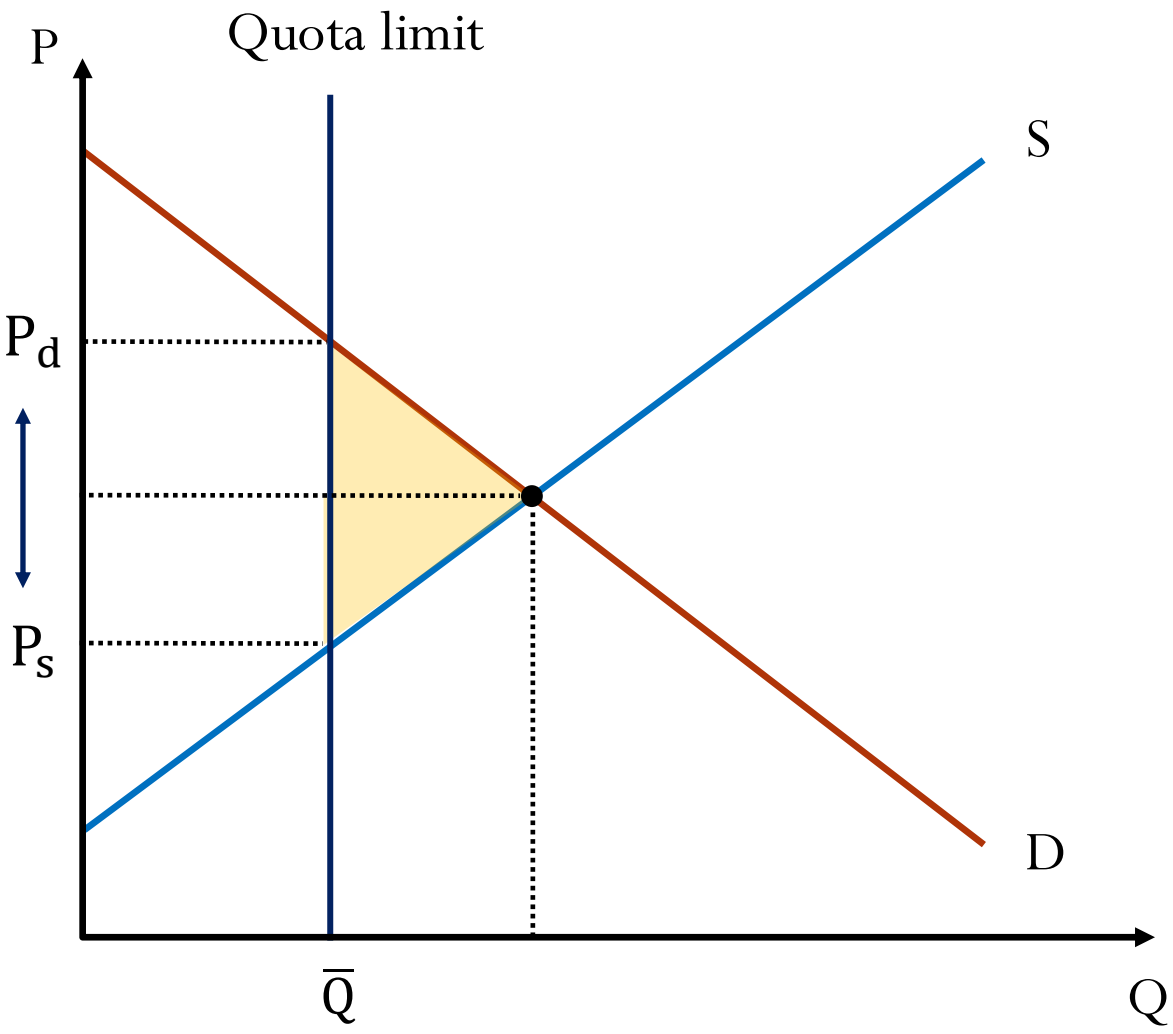
The price at which a specific quantity is supplied by producers.



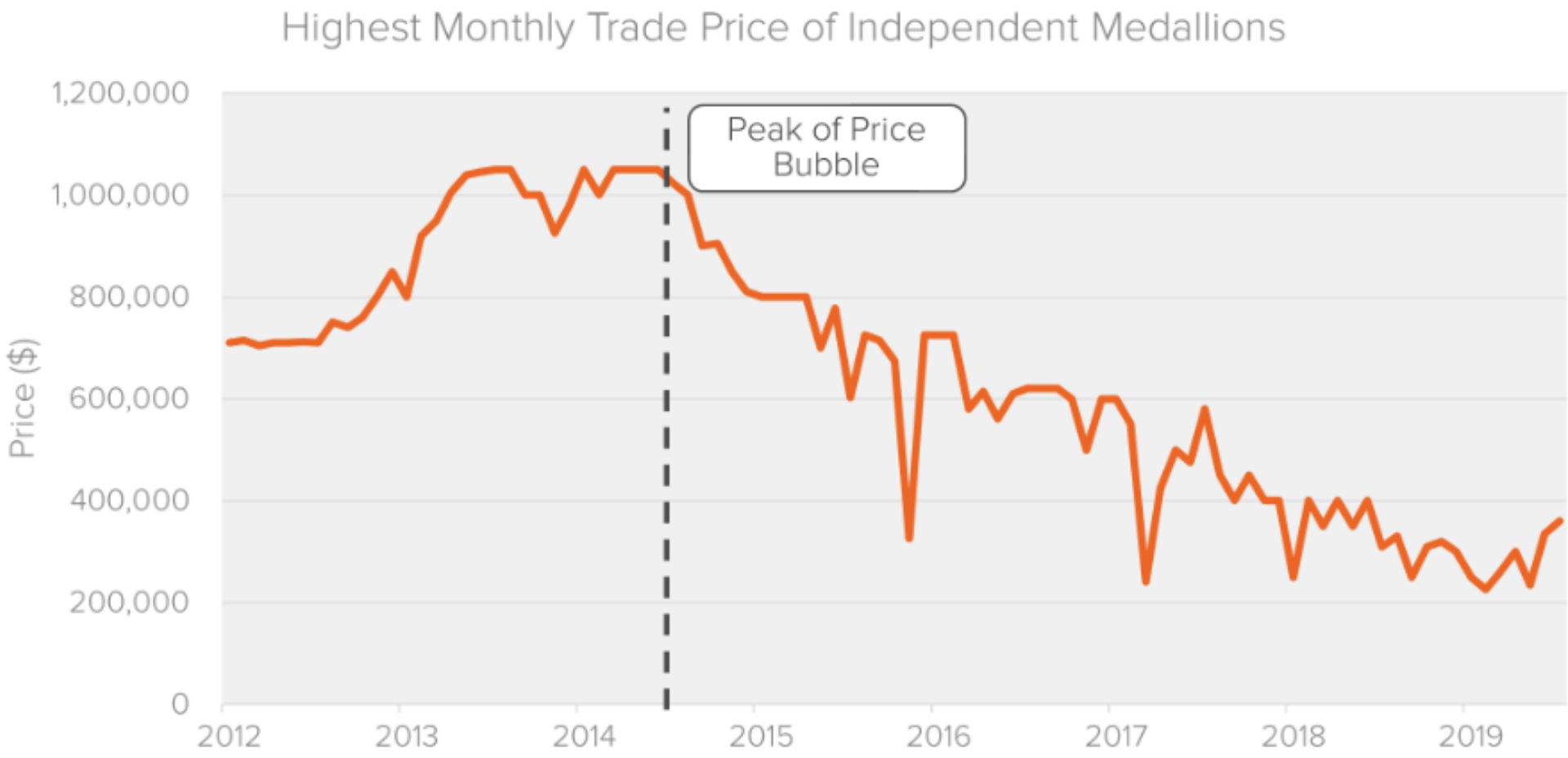
# Quantity Control

## Wedge (or Quota Rent)

The difference between the demand price and supply price at the quota limit.



# Quantity Control




*Source: New York City Taxi and Limousine Commission*

# How Quotas Cause Inefficiency?

- Inefficiently low quantity → deadweight loss
- Illegal activities (unlicensed taxis, smuggling, etc)
- Black markets

# Application: The 1973 Embargo

- Between 1973 and 1974, OPEC imposed an oil embargo on the US.
- This caused a sharp decrease in the supply of gasoline in the US domestic market.
- The government believed it was unfair to allocate gasoline only to consumers who are willing and able to pay high prices.
- To ensure "affordability", the government implemented a price ceiling on standard gasoline (57 cents per gallon).

# Long Lines at the Gas Station

- The series of events lead to people forming long lines at gas station.
- Who was responsible for the long gas lines?
- The OPEC oil embargo? Or the price ceiling implemented by the government?

# 20 Years Later

- 1981: Price controls on gasoline were abolished.

- Around 2000: Oil-producing countries jointly restricted output to push up price and achieved unexpected success.

Business

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## Opec cuts production to revive crude oil price

*By* **Dan Atkinson**

Wed 24 Mar 1999 03.26 GMT

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Scepticism greeted a long-awaited emergency cuts package from the 11-nation Organisation of Petroleum Exporting Countries (Opec) yesterday as investors took fright at the prospect of 'free riding' by non-Opec countries and cheating by cash-strapped cartel members. The key North Sea price, Brent crude for April delivery, fell 8 cents to $13.81 a barrel.

- Crude oil prices more than doubled, and gasoline prices at the pump soared.

# 20 Years Later

- In the US, people changed their driving and travel plans.
- The scenes of the 1970s were not repeated.
- In Europe, truck drivers and farmers blockaded oil shipments to protest high prices.
- In some European countries, severe gasoline shortages occurred.

## Autumn 2000: when fuel shortages brought UK to virtual standstill

Caroline Davies

**Rationing brings echoes of price protests that led to panic buying and filling stations running dry**

A driver takes part in the fuel price protest on 13 September 2000. Photograph: Ferran Paredes/Reuters

# Summary

- Price and quantity control interventions often cause inefficiency (relative to the equilibrium in a free market).
- But they may improve equity by affecting the distribution of surplus.
- In case of market failure, they may even improve efficiency.
- Next, Chapter 6: Elasticity.